

Roll No.

--	--	--	--	--	--	--

End Semester Examination 2025

Name of the Course	: B.Tech (ECE)	Semester	: V Sem
Name of the Paper	: Antenna and Wave Propagation	Paper Code	: TEC 504
Time	: 3 Hours	Maximum Marks	: 100

- Note**
- (i) All questions are compulsory
 - (ii) Answer any two sub questions among a, b & c in each main question
 - (iii) Total marks in each main question are twenty
 - (iv) Each question carries 10 marks

Q1

(10x2=20 marks)

- (a) Derive the Radar Range Equation and derive its simplified form assuming identical transmitting and receiving antennas. **CO 1**
- (b) Define and explain radiation pattern and its components. Discuss isotropic, omnidirectional, and directional radiation patterns with diagrams. **CO 1**
- (c) A center-fed dipole antenna has a length of 1.5 m and operates at a frequency of 100 MHz. Determine its electrical length in wavelengths and identify whether it behaves as a short or half-wave dipole. **CO 1**

Q2

(10x2=20 marks)

- (a) What are auxiliary potential functions? Derive the vector potential (A) equation for a given current distribution **J** in free space. **CO 2**
- (b) Derive the relationship between directivity and effective aperture of an antenna and state the conditions for maximum efficiency. **CO 2**
- (c) Two identical antennas with gains of 15 dB each are separated by 2 km in free space. The operating frequency is 5 GHz. Using the Friis transmission equation, find the ratio P_r/P_t . **CO 2**

Q3

(10x2=20 marks)

- (a) What is antenna array? Explain the principle of two-element array and derive its array factor. **CO 3**
- (b) Explain the advantages of antenna array. Compare broadside, end-fire, and scanning arrays. **CO 3**
- (c) Two isotropic antennas are spaced $\lambda/2$ apart and excited in phase. Determine the array factor and plot the radiation pattern. **CO 3**

Q4

(10x2=20 marks)

- (a) Discuss the geometry and working of a parabolic reflector antenna and derive its gain. **CO 4**
- (b) Discuss the different feeding methods for microstrip patch antennas. **CO 4**
- (c) A horn antenna operates at 8 GHz with an aperture area of 0.02 m². Compute its directivity. **CO 4**

Q5

(10x2=20 marks)

- (a) Explain free space loss. Derive the equation for line-of-sight (LOS) propagation **CO 5, 6**
- (b) Describe the structure of the ionosphere and its layers with typical electron densities. **CO 5, 6**
- (c) A free-space LOS microwave link operating at 10 GHz consists of a transmit and a receive antenna each having a gain of 25 dB. The distance between the two antennas is 30 km and the power radiated by the transmit antenna is 10 W. Calculate the path loss of the link and the received power. **CO 5, 6**

